

GENERIC REFINERY - GULF COAST

100 kBPSD Crude & Vacuum Distillation Unit

Basis of Design

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1 Introduction

This Basis of Design (BoD) defines the design premises for the 100 kBPSD Crude & Vacuum Distillation Unit (CDU/VDU). It is the governing input for every deliverable in the FEED package; all numbers below are read by the calculation code in cfu/basis.py, so this document and the calculations cannot diverge.

1.1 Scope

- Crude receipt from OSBL tankage, two-stage electrostatic desalting, crude preheat train (cold and hot trains).
- Atmospheric crude charge heater H-101 and atmospheric fractionator C-101 with three pumparounds and side strippers

for kerosene, diesel and AGO.

- Overhead condensing system, naphtha stabiliser (debutaniser) C-105 and naphtha splitter C-106.
- Vacuum heater H-201, wet packed vacuum column C-201, three-stage steam ejector system and hotwell.
- Unit utility distribution (steam, BFW, CW, fuel gas, nitrogen, instrument air), unit flare knock-out, closed drains.
- Excluded (interfaces only): tankage, sour water stripper, LPG treating, flare stack, wastewater treatment,

fuel gas production, central control room, main substation.

1.2 Capacity and operating envelope

Item	Value
Design capacity	100,000 BPSD (stream day)
On-stream factor	95% (8,322 h/y)
Turndown	50% of design
Hydraulic design margin	10% on normal flows (pumps, lines, control valves)
Run length	5 years between turnarounds

2 Feedstock

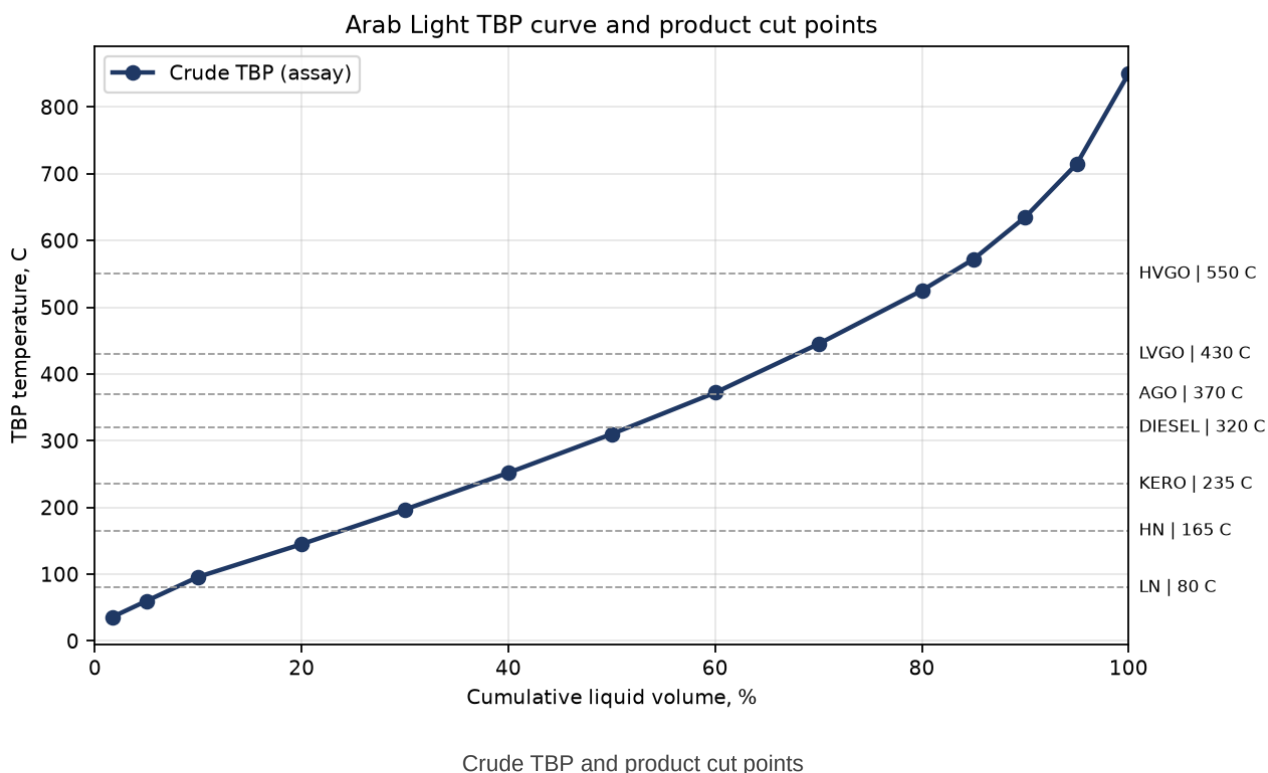
Property	Design crude	Check crude
Name	Arab Light (design)	Arab Heavy (check case)
API gravity	33.4	27.7
Sulfur, wt%	1.8	2.8
Salt as received, PTB	20.0	30
BS&W, vol%	0.3	0.5
TAN, mg KOH/g	0.05	0.1
Pour point, C	-30	-27

Check case: Hydraulic/thermal check case; not simulated in Rev A.

2.1 TBP distillation (design crude)

LV %	1.7	5	10	20	30	40	50	60	70	80	85	90	95	100
TBP C	36	60	96	145	197	252	310	372	445	525	572	635	715	850

Light ends (LV% on crude): C2 0.05, C3 0.4, iC4 0.25, nC4 1.0.



3 Products and specifications

Product	Specification / destination
LPG	C5+ <= 2 LV%; C2- <= 0.5 mol% (tie-in to LPG treating)
LN	TBP 36-80 degC; RVP <= 0.85 bar; to isomerisation
HN	TBP 80-165 degC; to NHT/reformer; ASTM D86 EP <= 185 degC
KERO	Flash >= 38 degC; D86 FBP <= 260 degC; to jet Merox/hydrotreater
DIESEL	D86 T95 <= 360 degC; flash >= 55 degC; to DHT
AGO	To DHT / FCC feed
LVGO	To hydrocracker; D1160 T95 <= 450 degC
HVGO	To FCC/hydrocracker; Ni+V <= 2 ppmw; CCR <= 0.8 wt%
VR	To delayed coker; 550 degC+ TBP

Atmospheric cut points (TBP): LN 80 C, HN 165 C, KERO 235 C, DIESEL 320 C, AGO 370 C. Vacuum cut points (TBP): LVGO 430 C, HVGO 550 C; VR = 550 C+. Desalted crude: salt <= 1 PTB, BS&W <= 0.2 vol%. Brine oil content <= 100 ppmw.

4 Site and climatic data

Item	Value
location	US Gulf Coast (generic)
elevation m	5
amb design C	35.0
amb min C	-5.0
wet bulb C	27.0
rel humidity	0.8
wind design	ASCE 7, 150 mph (3-s gust)
seismic	ASCE 7, SDC B
units	SI (degC, bar, kg/h, m)

Item	Value
pressure ref	bar(g) unless noted (a)

5 Utilities

Utility	Conditions
hp steam	P_barg=41.4, T_C=400, note=600 psig header
mp steam	P_barg=10.3, T_C=250, note=150 psig header
lp steam	P_barg=3.5, T_C=180, note=50 psig header
cooling water	supply_C=32, return_C=43, P_barg=4.5
bfw	T_C=120, P_barg=50
fuel gas	LHV_MJ_kg=47.0, P_barg=3.5, MW=20.0
instrument air	P_barg=7.0, dewpoint_C=-40
nitrogen	P_barg=7.0
power	utility_kV=13.8, mv_kV=4.16, lv_V=480, ac_hz=60, ups_V=120

6 Key process design parameters

Parameter	Value
atm top P	2.2
atm drum P	1.7
atm tray dP	0.008
atm cot max	370.0
atm overflash lv	0.03
atm drum T	45.0
tl dP	0.9
tl dT	5.0
pa split	{'TPA': 0.17, 'MPA': 0.27, 'BPA': 0.26}
steam lb per bbl	{'bottom': 10, 'KERO': 6, 'DIESEL': 6, 'AGO': 6}
vac top P	0.02
vac fz P	0.06
vac cot max	415.0
vac tl dT	12.0
vac overflash lv	0.03
vac steam lb per bbl	{'bottom': 5, 'coil': 1.5}
vac pa split	{'LVGO': 0.4, 'HVGO': 0.6}
stab drum P	11.5
stab drum T	45.0
split drum P	1.8
split drum T	50.0
h101 eff	0.9
h201 eff	0.84
rad flux kW m2	31.5
rad fraction	0.7
desalter T	135.0
wash water lv	0.05
min approach	20.0

7 Design criteria

- Design pressure: max(1.1 x max. operating, operating + 1.7 bar), minimum 3.5 barg; vacuum equipment full vacuum.
- Design temperature: maximum operating + 28 C (rounded up to 5 C).
- Corrosion allowance: 3 mm general, 6 mm for hot sulfidic (> 260 C) and overhead sour service.
- Materials: per API 939-C (McConomy curves) and API 571: 5Cr / 9Cr for sulfidic service above 260 C, 410S cladding for column shells above 260 C, Monel 400 for the C-101 top section (HCl/NH₄Cl dew point), Ti tubes in the overhead trim condenser, NACE MR0103 / HIC-resistant plate in wet H₂S service.
- Fired heaters: API 560; average radiant flux 31.5 kW/m² (H-101), 25 kW/m² (H-201); efficiency 90 % (LHV) with APH.
- Exchangers: TEMA R / API 660, max. 650 m² per shell, 20 C minimum approach, F >= 0.80.
- Pumps: API 610, 2 x 100 % (A/B), 10 % flow margin; spares on opposite electrical buses.
- Relief: API 520/521; all relief to closed flare header (hydrocarbon) via unit KO drum D-104.
- Instrumentation: ISA 5.1; DCS for regulatory control; SIS per IEC 61511 (SIL by LOPA); F&G separate.
- Electrical: NEC Art. 505 (Zone system), API RP 505 classification, IEEE 141/399 studies.

8 Codes and standards

Code	Application
API 650 / 620	Storage tanks (off-plot, interface only)
API 560	Fired heaters
API 660 / TEMA R	Shell & tube exchangers
API 661	Air-cooled heat exchangers
API 610 12th ed.	Centrifugal pumps
API 520 / 521	Pressure relief sizing and disposal
API 537	Flare details (interface)
ASME VIII Div.1	Pressure vessels and columns
ASME B31.3	Process piping
ASME B16.5 / B16.47	Flanges
API RP 505 / IEC 60079-10-1	Hazardous area classification (Zone system)
NFPA 70 (NEC) Art. 505	Electrical installation in Zone-classified areas
IEEE 141 / 399	Power system design and studies
ISA 5.1 / 5.4 / 5.2	Instrument symbols, loop diagrams, binary logic
IEC 61511 / ISA 84	Safety instrumented systems
IEC 62443	Industrial cybersecurity (zones and conduits)
API 2218 / 2510A	Fireproofing / LPG spacing
CCPS / GAP 2.5.2	Equipment spacing guidelines
NACE MR0103 / API 939-C / API 571	Materials for sour, sulfidation, naphthenic service

9 Environmental limits (design targets)

- Heater stacks: NO_x <= 25 ppmv @ 3 % O₂ (ultra-low-NO_x burners); SO₂ governed by fuel gas H₂S <= 160 ppmv.
- No continuous hydrocarbon venting to atmosphere; vacuum off-gas burned in H-201.
- Sour water to SWS; desalter brine to WWT via oil-recovery; noise <= 85 dB(A) at 1 m.